

GPS Collection Planning using Navmatix GPS Mission Planning

This assignment will require you to use the Navmatix GPS Mission Planning website:

<http://gnssmissionplanning.com/>.

Navigate to the Settings Tab. You will plan for a GNSS collection on 10/15/2019 at Latitude 39.5° N (entered as 39.5) and Longitude 77.3° W (entered as -77.3) between 8:00 AM and 8:00 PM. Set the cutoff value to 10° and the height to 0 meters. Set the time to UTC-5:00 Eastern Time (US & Canada). Make sure to click Apply to accept your settings.

In the Satellite Library Tab, turn off all of the constellations except GPS.

The screenshot shows the Navmatix website interface. The top navigation bar includes links for Home, Settings, Satellite library, DOPs, Number of Satellites, Elevation, Visibility, Sky Plot, and World View. The 'Settings' tab is active. On the left, there are input fields for Latitude (37.5), Longitude (-78.4), Height (0), Cutoff (10), Date (10/01/2017), From (08:00 AM), For (24h), and TimeZone ((UTC+01:00) Belgrade, Bratislava, Bl). A blue 'APPLY' button is at the bottom of the settings panel. On the right, there is a map of the Eastern United States with a red dot indicating the 'Current position' near Roanoke, Virginia. Below the map is a search bar labeled 'Search address' and a 'Search results' dropdown menu. The footer of the page reads '© 2017 Navmatix s.r.o.'

Type up a short report that addresses the following.

1. Are there any GPS satellites in the NAVSTAR constellation that are currently considered “unhealthy”? (2 Points)
2. Are there any GLONASS satellites that are currently considered “unhealthy”? (2 Points)
3. Are there any Galileo satellites that are currently considered “unhealthy”? (2 Points)
4. Copy the Number of Satellites graph to your report. (3 Points)
5. What is the maximum number of NAVSTAR (labelled as GPS) satellites that are predicted to be detected during this time period? (2 Points)
6. What is the minimum number of NAVSTAR (labelled as GPS) satellites that are predicted to be detected during this time period? (2 Points)

7. Copy the DOPs graph to your document. (3 Points)
8. At what time, roughly, is the largest PDOP (positional) spike? (3 Points)
9. Make a copy of the Visibility graph to your document. (3 Points)
10. On this day and at this location, when will satellite G15 be observable? (3 Points)